LOGO UNIVERSITY

PROJECT NAME

Term Project Report

Dynamic Hand Gesture Recognition
using Bag-Of-Visual-Words

Student: Name, ID

Lectuer: Teach Name

Date

1 Introduction

Hand gesture recognition has been rapidly developed in the human-machine interaction applications in recent years. Firstly, hand-gestures are intuitive and effective in expressing human feelings. Second, the development of sensor technology has brought hand-gesture such as sensors using accelerometers or gyos to capture accurately the movement of the hand and fingertips, multi-touch screen sensors widely available through tablets, telephone devices, and visual-based sensors for hand-recognition through color images [1].

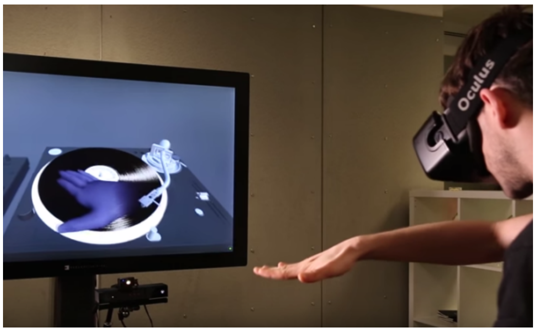


Figure 1: Human-machine interaction applications

In recent years, thanks to developing quickly 3D depth sensors with low-cost such as Microsoft Kinect, Intel RealSense, it brings many benefits in dealing with hand-gesture recognition more than traditional sensors. Firstly, it is robust to light variants, background clutter, and occlusions. So, it helps easily in hand detection and segmentation. Secondly, the depth sensors capture 3D information in the scene. It helps the development quickly of hand-pose, human-pose estimation in determination the skeleton of human body or hand. Therefore, there are many choices for getting information in hand-gesture recognition such as depth, color images, and body/hand skeleton [2].

There are two main categories in hand gesture recognition: static and dynamic hand gesture recognition. Different from static hand gesture recognition detecting hand region and extracting hand feature from hand segmentation at the specific time, the dynamic hand gesture recognition needs to exploit more the temporal features from the hand shape sequences. It treats as the pattern recognition problems consisting of feature extraction, and classification.

Up to now, the dynamic hand gesture recognition is a challenging task due to its small size, complexity, and self-occlusion. Moreover, it is difficult to recognize because of intra-class dissimilarities, inter-class similarities in gestures. Intra-class gesture dissimilarities come from cultural or individual factors to lead the differences of position, speed, and style of the hand gesture. Inter-class similarities appear in high same among some hand gestures. So, it needs to deal with exploiting the spatial and temporal information of hand gestures to prevent above problems as well as the noise from the device.

In this project, we propose the method for dynamic hand gesture recognition based on exploiting the depth and skeleton features. For depth features, we use SURF to extracting visual code-words. With skeleton features, we use 2D skeleton information to calculate the distance of all combinations from two joints. All sparse features will be clustered with Gaussian Mixture Models (GMM), and k-means to find the specific visual words. From there, we build bag-of-visual words for all gestures. To exploit temporal model, we use temporal pyramid matching on every scale to build features for gesture sequence. Finally, we use Supper Vector Machine (SVM) to classify the sequence gesture. We evaluate our method on DHG Dataset [3] with late fusion techniques achieving the good accuracy.

2 Related Research and Key Technology

2.1 Dynamic Hand Gesture Recognition

With the rapid development of hand pose estimation [4] and supported from depth-based cameras such as Intel RealSense, Microsoft Kinect [5], the hand skeleton features are interesting in the hand gesture recognition in recent works.

Lu et al. [6] use the palm direction, palm normal, fingertips positions and palm center position data from Leap Motion controller to extract features such as fingertip-distances, fingertip-angles, fingertip-elevations, adjacent fingertip-angles for dynamic hand gesture recognition. Garcia et al. [7] collected RGB-D sequences as well as hand-pose annotation for first-person hand action recognition. The best base-line method is in merging color, depth and pose data. A multi-modal deep learning framework proposed by Neverova et al. [8] uses color, depth, audio stream as well as body skeleton. The final label of a sequence is computed from voting every frame.

The most recent works, De Smedt et al. [3] published DHG with depth and 2D/3D skeleton information to deal with the lack of benchmark and comparison methods in dynamic hand gesture in depth and 3d hand joint approach. They introduce Shape of Connected Joints (SoCJ) descriptor to represent hand shape. After that, Fisher Vector computed from SoCJ descriptor, as well as histograms of the hand direction and the wrist orientation are used for classification. Their method also is the state-of-the-art in handcrafted methods such as HOG2, HON4D, etc.

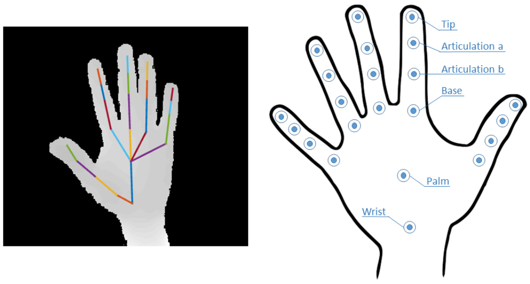
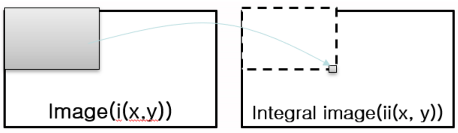


Figure 2: Hand gesture annotations in DHG Dataset

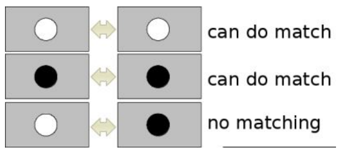
2.2 Local Feature Descriptors

The Scale-Invariant Feature Transform (SIFT) [9] includes both a detector and a descriptor. It was designed to be scale and rotation invariant which provides it with robustness against affine image transformations and, hence, repeatability. SIFT uses a multi-scale scheme so that interest points at different scales can be found.

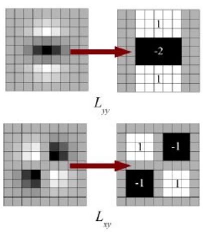
SIFT was comparatively slow and needed more speeded-up version. Speeded Up Robust Features (SURF) [10] was introduced a new algorithm for speeded-up version of SIFT as Fig.3. (1) SURF goes a little further and approximates Laplacian of Gaussian (LoG) with Box Filter. One big advantage of this approximation is that, convolution with box filter can be easily calculated with the help of integral images. And it can be done in parallel for different scales. (2) For orientation assignment and feature description, SURF uses wavelet responses in horizontal and vertical direction. Wavelet response can be found out using integral images very easily at any scale. (3) The sign of Laplacian (Trace of Hessian Matrix) is used for underlying interest point. It adds no computation cost since it is already computed during detection. The sign of the Laplacian distinguishes bright blobs on dark backgrounds from the reverse situation. In the matching stage, we only compare features if they have the same type of contrast, which allows for faster matching, without reducing the descriptor's performance.



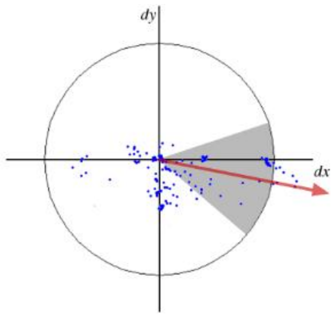
(a) Integral Image



(b) Matching descriptor using sign Laplacian



(c) Box filter approximation Laplacian of Gaussian



(d) Wavelet response in feature descriptor

Figure 3: Speeded Up Robust Features (SURF)

2.3 Bag-of-Visual Words

The set of SIFT or SURF descriptors of the images in the training set are clustered using k-means, for a user-provided value of k, and the centroids of each cluster represent the visual words of our vocabulary. To compute these clusters and their corresponding centroids, k-means starts by some (random) initial centroids and then proceeds until convergence. Then, once this vocabulary is built, a given new local descriptor is assigned to the word corresponding to its closest centroid. Therefore, an image in both the training and test sets is coded as a k-bin histogram (the bag of words) with the counts of the words of all its descriptors. Fig.4 shows the common process of bag-of-visual words in object recognition. [11]

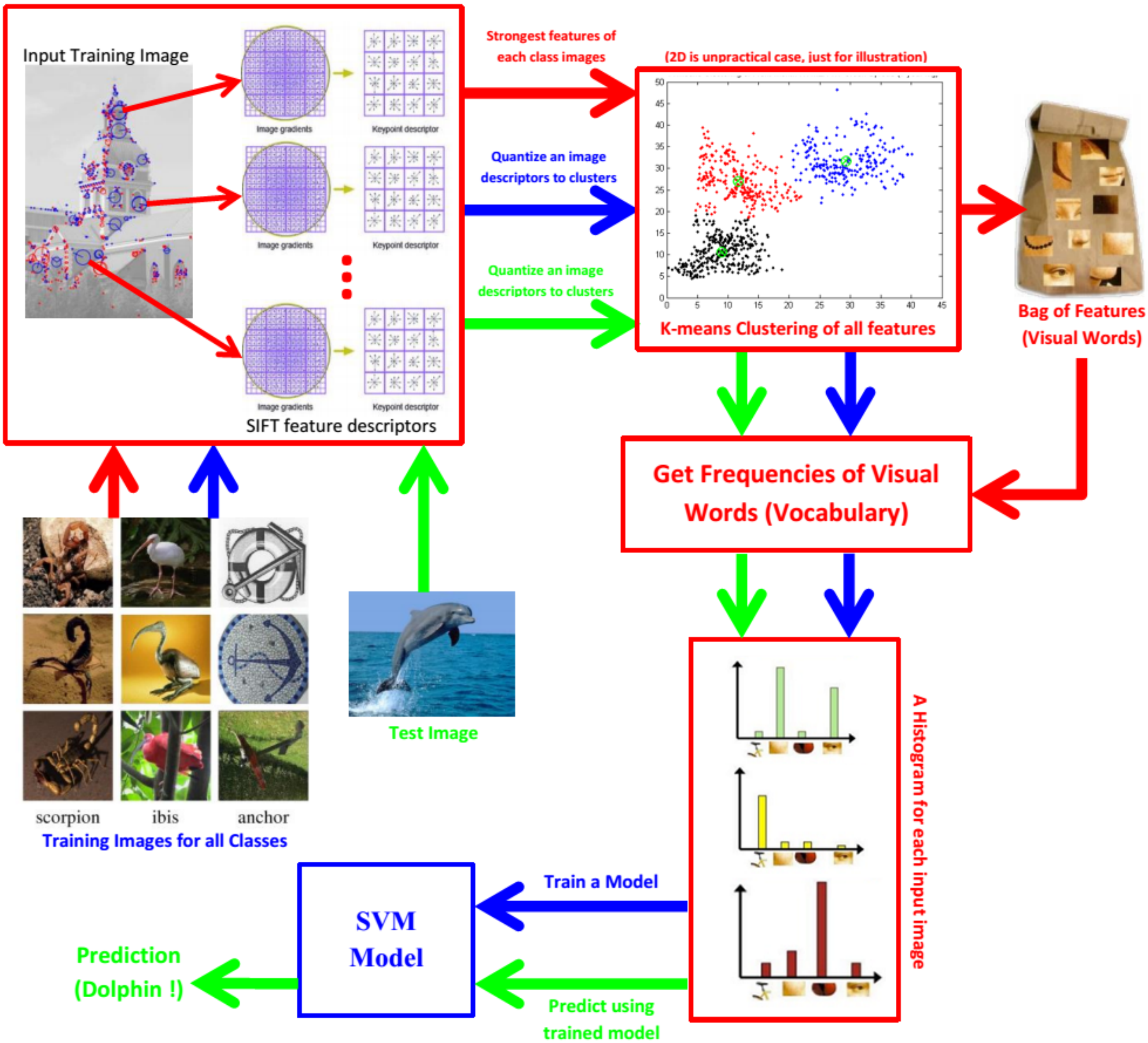


Figure 4: Bag-of-Visual-Words process in object recognition problem

2.4 Temporal Model

The dynamic hand gesture recognition task involves modeling the temporal aspect of gestures in addition to the feature extraction. The literature of dynamic hand gesture temporal modeling shows two distinct strategies: (1) Creating descriptors which carry spatial and temporal information; (2) Modeling sequences of spatial descriptors via temporal classifiers.

Spatio-temporal descriptors are widely used to recognize gestures from videos. Several of these descriptors have evolved from their 2D versions, by augmenting existing descriptors with additional features extracted along the temporal dimension. For example, Ohn et al. [12] introduced a 3D version of the HOG descriptor, called HOG2, to handle videos and Klaser et al. [13] proposed similarly the HOG3D descriptor.

As characterizing a whole hand gesture sequence using a single representation can lead to miss-classification due to inverse gesture, they used a temporal pyramid as Fig.5. The principle of the temporal pyramid is to divide a sequence into n sub-sequences at each n^{th} level of the pyramid. The final representation is the concatenation of all spatio-temporal descriptors computed on all sub-sequences. This strategy allows to distinguish the beginning, the middle and the end of gestures.

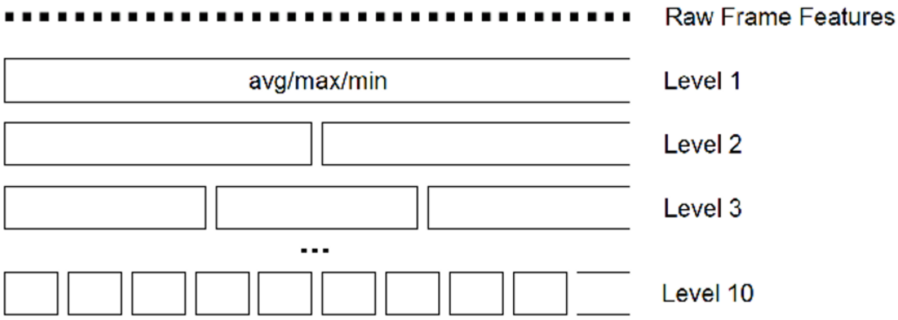


Figure 5: Temporal pyramid model

Sequence modeling uses statistical algorithms that take a sequence as input. In the category of sequence modeling methods, Hidden Markov Models (HMMs), Hidden Conditional Random Fields (HCRFs) and their extensions have proven to be efficient in many sequential recognition tasks. While HMM models joint probability of states and observed features, the parameters in HCRFs model are conditioned only on observations. [14]

3 Project Contents and Algorithm

3.1 Overview

In this scope of subject, we proposed dynamic hand gesture recognition to exploit the appearance changes of hand shapes by temporal information. Fig. 6 shows our proposed methods with 5 main steps.

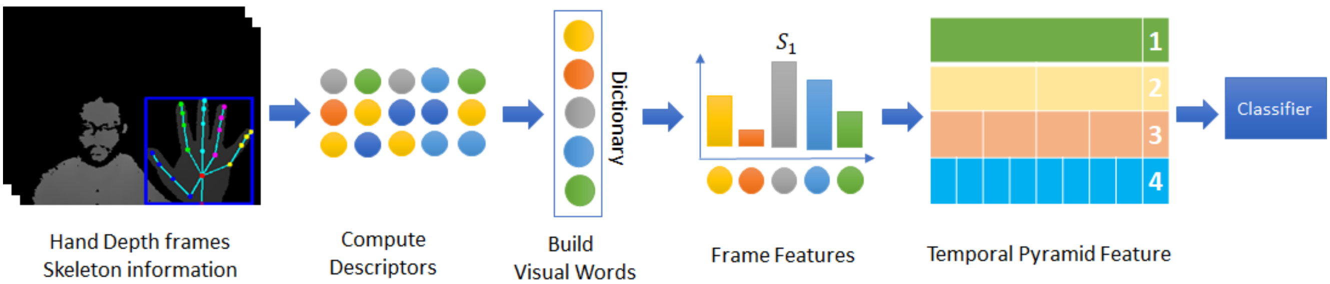


Figure 6: Proposed method

In first step, we will extract hand shape features from depth image and skeleton information from every frame of sequences in the dataset. For hand depth image, we use SURF descriptors. With skeleton information, we calculate from distance of all finger joint pairs as well as normalization. Next step, we build visual words using two methods: k-means clustering and Gaussian Mixture Model clustering. In third step, we will build frame-level features by histogram of visual words. Finally, sequence features will merge from temporal pyramid of 4-level sub-sequences. The SVM classifier will be used to recognize the specific gesture of sequence.

3.2 DHG Dataset

In this project, we use DHG dataset with 14 classes as Table 1 is applied in dynamic hand gesture recognition.

Table 1: Gesture List in DHG Dataset

Labels	Gestures
1	Grab
2	Tap
3	Expand
4	Pinch
5	Rotation CW
6	Rotation CCW
7	Swipe Right
8	Swipe Left
9	Swipe Up
10	Swipe Down
11	Swipe X
12	Swipe V
13	Swipe +
14	Shake

DHG dataset has 2800 sequences with 20 participants for 5 trials in 2 ways depending on the number of fingers with one finger and the whole hand. The depth images and hand skeletons were received from Intel RealSense camera. We also split DHG dataset with 70% for training and 30% for validating.

We only use depth image and skeleton information in 2-dimesion space to evaluate the proposed method. Skeleton information consists of wrist, palm and 5 finger joints with 4 joints per finger.

3.3 Hand Depth Features

In every depth frame receiving from camera, it contains depth images and skeleton information with finger joints as in Fig. 7.

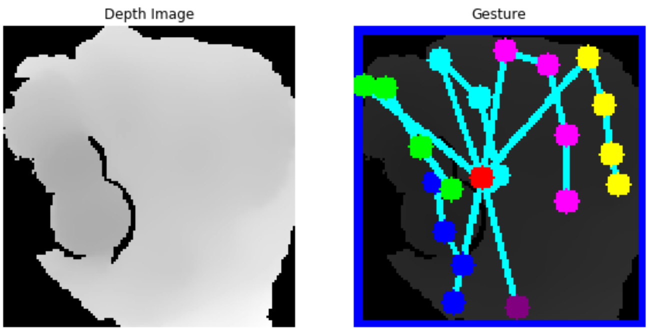


Figure 7: Hand information

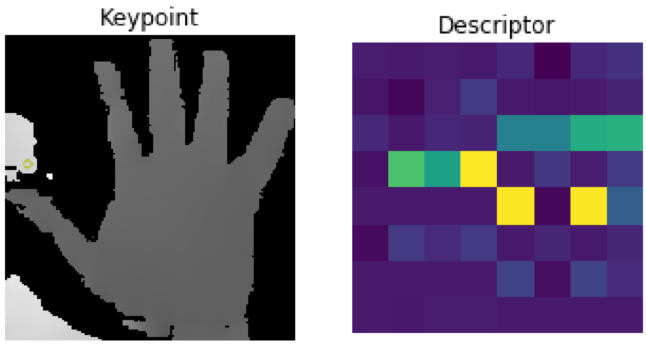
In hand depth features, we use SURF algorithm to extract the key-points in 64 dimension as Fig.8

We combines all SURF descriptors into local patches. Gaussian Mixture Model and k-means clustering helps to vote the k-center visual words from all patches. In this project, k has value 500 to be suitable the size when applying the temporal pyramid model. With 4-level temporal pyramid, there are 7500 dimension per sequence feature vector as in Fig.9. Before training, we use min-max scale to normalize the features.

3.4 Hand Skeleton Features

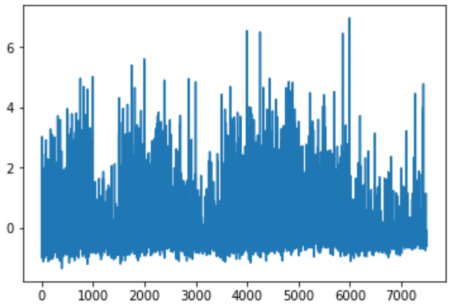
For hand skeleton features, all distinct joint pairs will be calculated the distance as in Fig.10a. It contains the tuples of hand joints representing the hand's physical structure as presented in Fig.10b. As hand depth feature

(a) Key-points

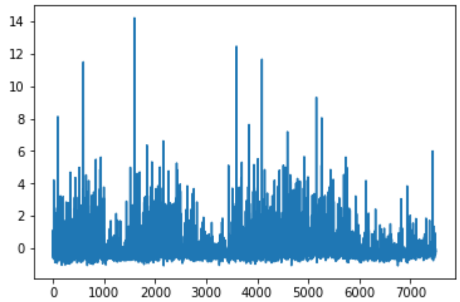


(b) Descriptor

Figure 8: SURF features in one frame



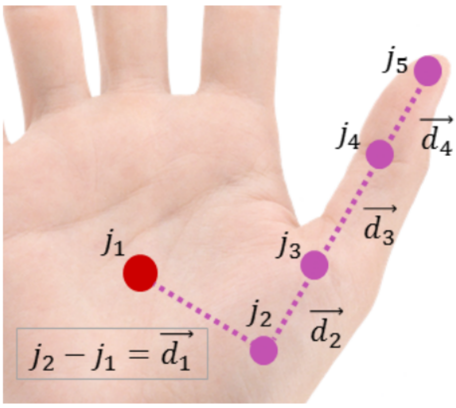
(a) k-means



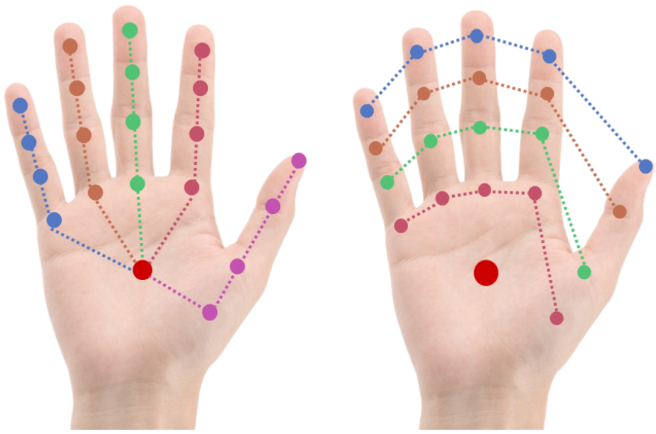
(b) Gaussian Mixture Model

Figure 9: Temporal pyramid hand depth sequence feature

process, clustering algorithms will create k visual-words. Every hand skeleton feature belongs the specific visual word. All features in an sequence are divided into n-level sub-sequences to create temporal pyramid feature for the dynamic hand gesture of the sequence.



(a) Finger-joint distance



(b) An examples of nine pairs intuitively for descriptors

Figure 10: Skeleton Features

4 Experiment and Result

4.1 Environments

We built the program on Window environment with Python 3.5. In the program, we used skimage, sklearn, and opencv as main libraries to developlearning models. The experiment was done with the machine's configurations as follows: Intel(R) Core(TM) i7 -8700 CPU @3.20 GHz, GTX 1080Ti 11GB RAM.

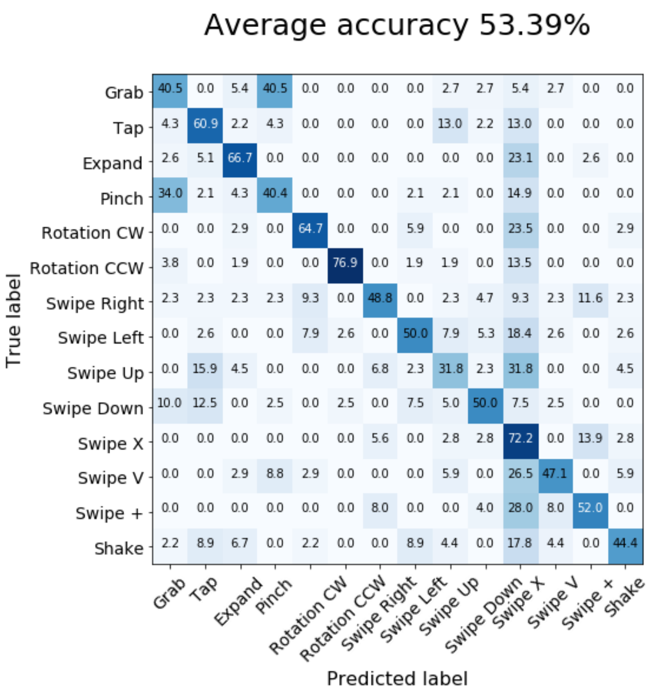
4.2 Training Settings

In training process, we choose Gauss Mixture Model and k-means algorithm to create the visual words. With limited time and memory, the number of clusters has value 500. The number of patches for Gauss Mixture Model is 100 x clusters with randomly choosing. It takes 4 hours for fitting on training data.

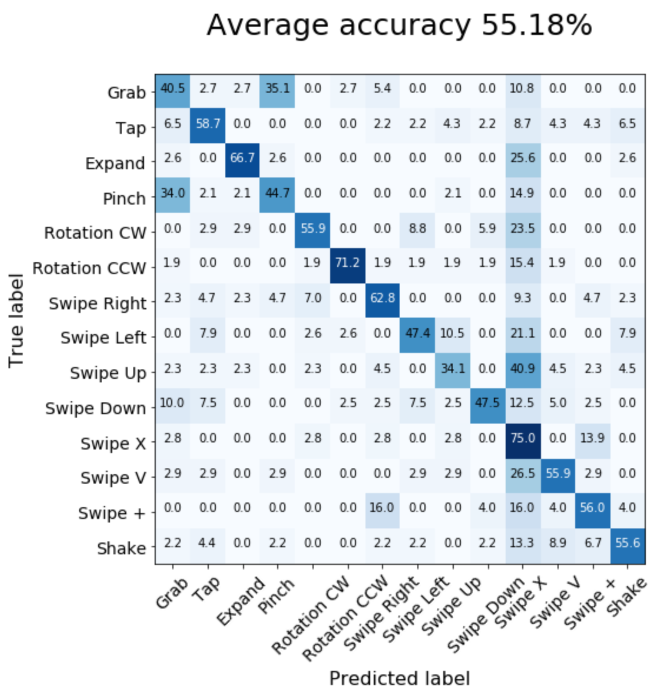
SVM classifier has the regular parameter C 100, gamma 'auto', and kernel 'rbf'. Besides, we set probability parameter True to produce the probability of prediction applied in fusion process. The training data will be normalized with min-max scaler.

4.3 Skeleton Features

Fig.11 shows results of our methods with skeleton features.



(a) Kmeans



(b) Gaussian Mixture Model

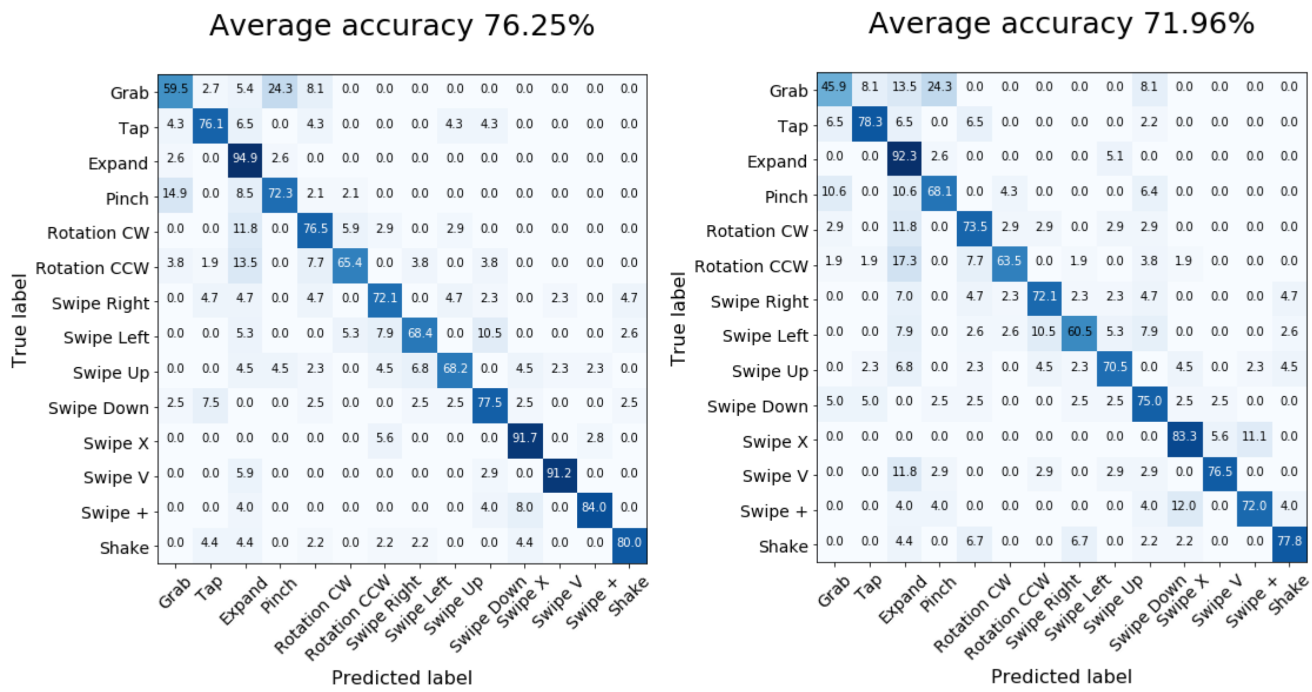
Figure 11: Confusion Matrix on skeleton features

Accuracy of skeleton features are 53.39%, and 55.8% for k-means and Gaussian Mixture Model (GMM), respectively. GMM in this case is better than k-means because it is not over-fitted by only choosing a subset of training data in clustering. Moreover, it is not enough information for k-means to cluster the data, which leads to stop the clustering process early.

With difficulty gestures such as Grab, Pinch, etc., they focus on the motion and keep hand shape. So, the accuracy in these gestures are not good. In future, we need to encode the motion and pose feature based on palm and wrist joints among successive frames.

4.4 Depth Features

Fig.12 shows the confusion matrix result of recognizing on hand depth features.



(a) Kmeans

(b) Gaussian Mixture Model

Figure 12: Confusion Matrix on depth features

The reason of enhancement comparing on skeleton features is from depth information. It contains not only hand shape but also motion information. SURF catches up the key-points on changes from motion and appearance changes. So, the accuracy incrases from 55.18% into 76.25%. Moreover, in during clustering process, k-means use all training data and achieve good converging. Therefore, it has the accuracy higher than Gaussian Mixture Model only using the sub-set of data.

4.5 Fusion

To enhance the performance of the two models, we use fusion techniques, which exploit the complement and redundancy information between the models. In this paper, we use late fusion as described in [15]. In late fusion, we use maximum, average and weighted operator on four models using skeleton, depth models with Gaussian mixture model, and k-means clustering.

The late fusion technique combines the probability outputs of each model by majority voting. Given $\hat{y}_1$, $\hat{y}_2$, $\hat{y}_3$ and $\hat{y}_4$, respectively, the output probabilities of skeleton gaussian, skeleton kmeans, depth gaussian, and depth kmeans models, the final predicted labels are computed as below equation:

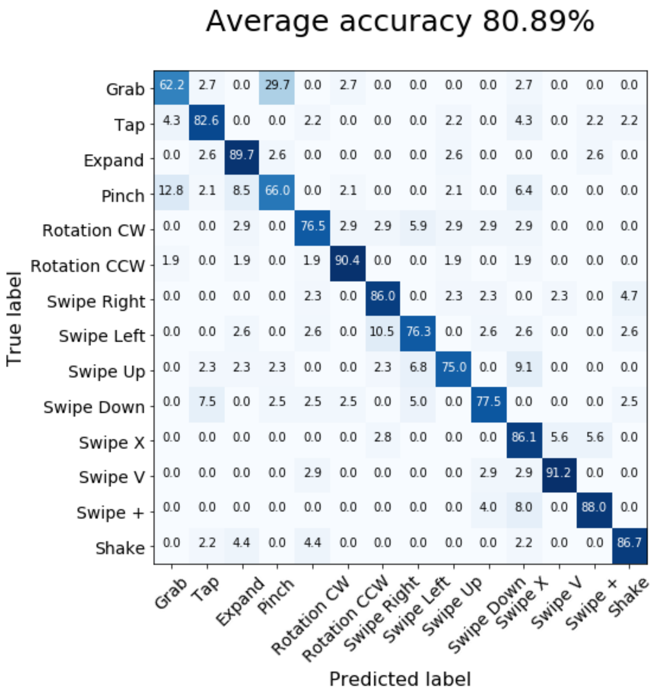
$$\hat{y}_{maximum} = arg\max\left\{\max_{i=1}^{4}\hat{y}_i\right\} \tag{1}$$

$$\hat{y}_{avg} = arg\max\left\{\left(\sum_{i=1}^{4}\hat{y}_i\right)/4.0\right\} \tag{2}$$

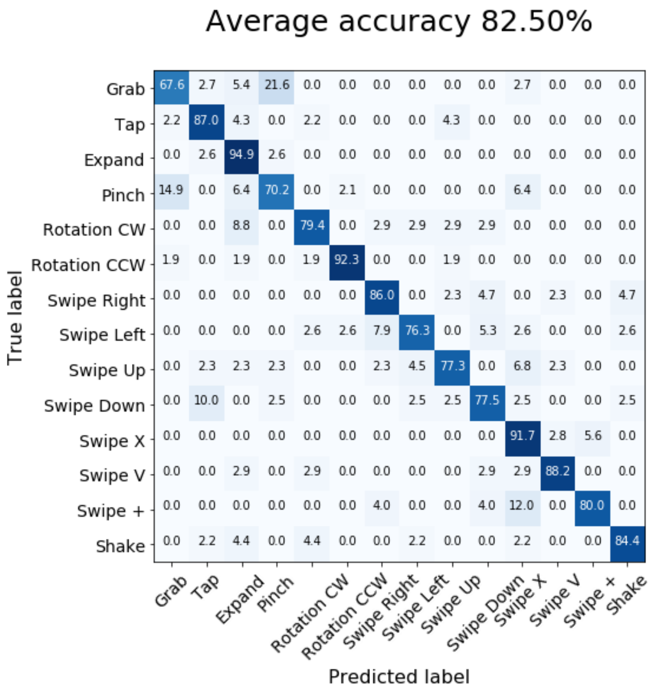
$$\hat{y}_{weighted} = \sum_{i=1}^{4} w_i.y_i with \sum_{i=1}^{4} w_i = 1 \tag{3}$$

where $w_i \in [0, 1]$ is the parameter depending on the performance of each network. In practice, we adjust from zero to one with step size 0.01 to find the optimal value of the classification result.

Fig. 13 shows the confusion matrix by combining four models by maximum, average operator. With fusion technique, the accuracy increase from 71% in depth model, and 55% in skeleton into 82.50% by integration skeleton and depth features from Bag-Of-Visual-Words models.



(a) Maximum Fusion



(b) Average Fusion

Figure 13: Confusion Matrix on fusion models

Moreover, by weighted sum technique, we try to search optimal weighted values in every model and increase accuracy from 82.50% to 83.75% as in Fig.14. The weighted parameters for best accuracy is $0.24, 0.1, 0.29, 0.37$ for skeleton gaussian mixture, skeleton kmeans, depth gaussian mixture, depth kmeans models, respectively.

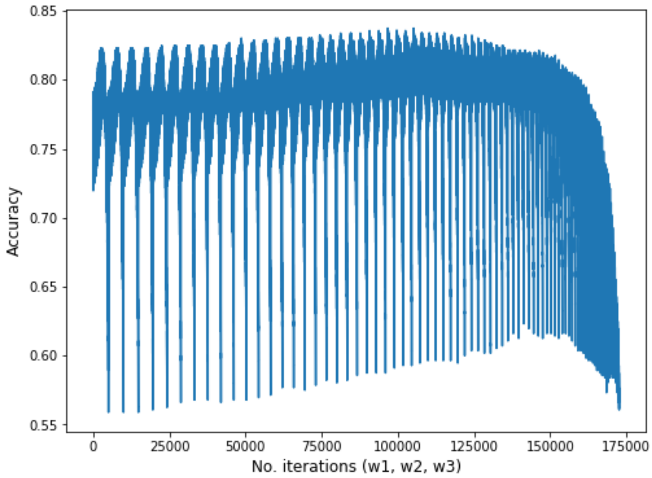
Table 2 shows the results of our proposed methods in this project:

Table 2: Summary of results in our proposed methods

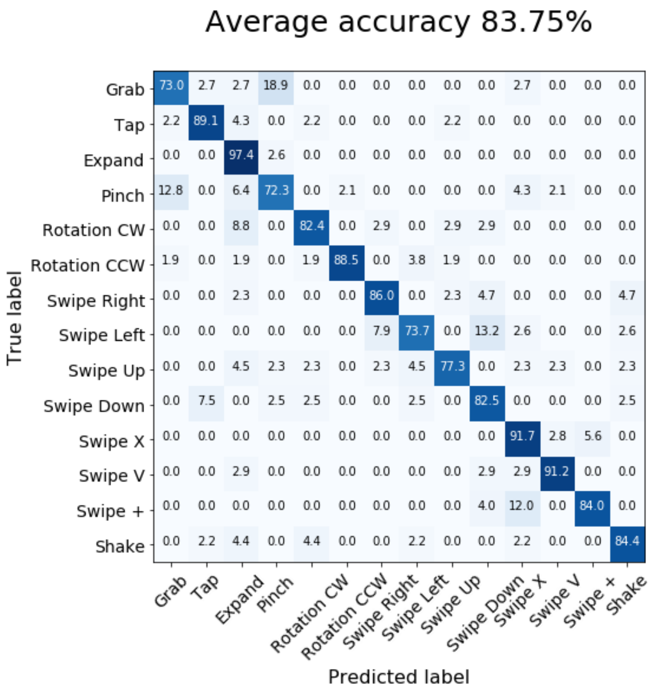
Method	**Features**	**Clustering**	**Accuracy**
1	Skeleton	k-means	53.39%
2	Skeleton	GMM	55.18%
3	Depth	k-means	76.25%
4	Depth	GMM	71.96%
5	Fusion	Max	80.89%
6	Fusion	Average	82.50%
7	Fusion	Weighted	83.75%

5 Conclusions

In summary, we propose a dynamic hand gesture recognition method with depth and skeleton classification approach based on the hand-shape feature extraction and fusion between hand-shape model and hand-skeleton model. Our experimental results achieve good accuracy corroborate that the hand-shape features can cope with various complexity, low-resolution and self-occlusion of hand shape changes in the gestures. Furthermore, we also demonstrate that our method achieves the best accuracy on DHG dataset with the traditionally handcrafted methods.



(a) Search parameters w_1, w_2, w_3, w_4



(b) Confusion Matrix

Figure 14: Weighted fusion Model

References

[1] S. S. Rautaray and A. Agrawal, “Vision based hand gesture recognition for human computer interaction: a survey,” *Artificial intelligence review*, vol. 43, no. 1, pp. 1–54, 2015.

[2] J. Wang, Y. Chen, S. Hao, X. Peng, and L. Hu, “Deep learning for sensor-based activity recognition: A survey,” *Pattern Recognition Letters*, vol. 119, pp. 3–11, 2019.

[3] Q. De Smedt, H. Wannous, and J.-P. Vandeborre, “Skeleton-based dynamic hand gesture recognition,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, pp. 1–9, 2016.

[4] J. S. Supancic, G. Rogez, Y. Yang, J. Shotton, and D. Ramanan, “Depth-Based Hand Pose Estimation: Data, Methods, and Challenges,” in *2015 IEEE International Conference on Computer Vision (ICCV)*, 2015.

[5] L. A. Anonymous, E. Krupka, N. Bloom, D. Freedman, A. Vinnikov, and A. B. Hillel, “Toward realistic hands gesture interface : Keeping it simple for developers and machines,” *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems - CHI ’17*, pp. 1887–1898, 2017.

[6] W. Lu, Z. Tong, and J. Chu, “Dynamic hand gesture recognition with leap motion controller,” *IEEE Signal Processing Letters*, vol. 23, no. 9, pp. 1188–1192, 2016.

[7] G. Garcia-Hernando, S. Yuan, S. Baek, and T.-K. Kim, “First-person hand action benchmark with RGB-D videos and 3D hand pose annotations,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 409–419, 2018.

[8] N. Neverova, C. Wolf, G. Taylor, and F. Nebout, “Moddrop: adaptive multi-modal gesture recognition,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 38, no. 8, pp. 1692–1706, 2015.

[9] D. G. Lowe, “Distinctive image features from scale-invariant keypoints,” *International journal of computer vision*, vol. 60, no. 2, pp. 91–110, 2004.

[10] H. Bay, T. Tuytelaars, and L. Van Gool, “Surf: Speeded up robust features,” in *European conference on computer vision*, pp. 404–417, Springer, 2006.

[11] L. Fei-Fei and P. Perona, “A bayesian hierarchical model for learning natural scene categories,” in *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'05)*, vol. 2, pp. 524–531, IEEE, 2005.

[12] E. Ohn-Bar and M. Trivedi, “Joint angles similarities and HOG2 for action recognition,” in *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pp. 465–470, 2013.

[13] A. Klaser, M. Marszałek, and C. Schmid, “A spatio-temporal descriptor based on 3d-gradients,” in *BMVC 2008-19th British Machine Vision Conference*, pp. 271–275, British Machine Vision Association, 2008.

[14] S. B. Wang, A. Quattoni, L.-P. Morency, D. Demirdjian, and T. Darrell, “Hidden conditional random fields for gesture recognition,” in *2006 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'06)*, vol. 2, pp. 1521–1527, IEEE, 2006.

[15] D. Wu, L. Pigou, P. J. Kindermans, N. D. H. Le, L. Shao, J. Dambre, and J. M. Odobez, “Deep Dynamic Neural Networks for Multimodal Gesture Segmentation and Recognition,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 38, no. 8, pp. 1583–1597, 2016.